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Player-centered game AI from a flow perspective: Towards a better understanding of past trends and future directions

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ABSTRACT

Player-centered approaches that aim to maximize player enjoyment have been steady, but with poor heuristics that do not rely on any particular theory of entertainment. Certainly, the Theory of Flow is the most referred in the game AI area and, still, it is unclear how to effectively design and implement adaptive game modules or understanding which game features drive players to a flow state. Therefore, in this document we perform a systematic analysis of literature aimed to enhance our knowledge about how to adapt flow heuristics to a video game context. This analysis endows us with the knowledge needed to define a flow framework for game AI. Our framework, FlowAI, describes which modules and what gameplay features can be adapted to design an effective video game intended to facilitate the achievement of flow in players. Furthermore, from our framework standpoint, we identify and review existing work that could be adjusted to foster flow. Our aim with this analysis is to identify current challenges, and motivate new directions in the player-centered research area.
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1. Introduction

In the beginning, the Game AI area was primarily focused on using video games as a testbed for artificial intelligence algorithms. Nowadays, there are many approaches that have the goal of maximizing player enjoyment through artificial intelligence techniques [1–4]. All these works can be classified in the player-centered subarea of Game AI [5]. The basic idea of these approaches is to make games reactive to players by adapting key game features, such as: space adaptation, missions, character behaviors and difficulty scaling. While each game feature is different to each other, their adaptation mechanisms share an essential module to create a personalized experience: a player model.

The objectives of a player model are to define the player's preferences – towards specific gameplay features – and guide the adaptivity modules about – how and when – they should react, in order to maximize the player's enjoyment. Despite significant advances in this subject, most of the proposed heuristics to adapt game components are vaguely based on theories of entertainment [6]. This lack of formalism ignores decades of research about user experience in other areas.

The Theory of Flow is a well-established theory of entertainment. This theory focuses on learning which are the precursors and outcomes of the optimal experience called flow [7]. When a subject gets into flow, she experiences an intense level of attention [7]. Also, this subjective state of mind has been described as a deep immersion state [8]. Nevertheless, it is unclear how the concepts of flow and immersion are related to each other [9–11].

There have been many efforts to adapt the flow theory to many contexts, including video games [4,6,12–16]. However, it is still unclear how to effectively design and implement adaptive game modules, or even to discern which game features drive players to a flow state. Nevertheless, there is theoretical and empirical evidence indicating that a perfect balance between the challenge faced by the player and her skills is vital if we want to avoid a short-lived flow experience [9,17,18]. The Dynamic Difficulty Scaling (DDS) subarea, is arguably one of the most studied and yet, it is a field with much room for improvement, from a flow standpoint.

This lack of formalism is natural, due to the immaturity of the research area under analysis. As a first step towards a game AI field with better theoretical foundations, we performed an exhaustive analysis on approaches aimed to adapt flow elements to a video game context. Next, we elaborated our approach towards immersion and how this concept contributes in the design of better flow experiences. These analyses endows us with knowledge to understand which game AI subareas can contribute to foster flow.

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From this basis, we present a flow framework that unifies key elements from preexisting approaches, which are suitable for a game AI context. We structured our framework, FlowAI, as a generic player-centered game. Besides, we detailed the relation between its modules in a functional manner.

Accordingly, from our framework standpoint, we proceed to survey the game AI subareas that can contribute to foster flow. Hence, the survey is separated in three parts: player modeling, dynamic difficulty scaling and immersion. We are focused on learning about current trends and challenges. In addition, we believe this work will motivate new directions in the player-centered research area.

2. Overall approach

First, in Section 3 we give a brief introduction to the Theory of Flow. Here, we introduce the constituent concepts to understand flow and how this concept has evolved over the years. Then, in Section 4 we present an exhaustive analysis on approaches aimed to adapt flow elements to a video game context. Specifically, this comprehensive review is aimed to determine the elements of flow that are particularly appropriate to design player-centered games proposed to foster flow. Also, we identify the main open questions in this area. One of those questions is how the concepts of immersion and flow are related to each other. Thus, in Section 5 we argue our standpoint on flow and immersion.

With this theoretical foundation, we present in Section 6 a flow framework for game AI. This framework, FlowAI, unifies key elements from preexisting approaches that better suit a game AI context. Further, in contrast to others, it is detailed in a lower level of abstraction; with the structure of a generic player-centered game. Because its modules can be interpreted as game AI subareas that can contribute to design better games that facilitate, for all kind of players, getting into a flow state. The major game AI subareas our framework includes are: player modeling, dynamic difficulty scaling and immersion. Therefore, through the lenses of our framework, we proceed to survey these subareas in Sections 7–9 respectively. The format of those sections is shared. First, an overview of the state of the art is given. Then, we criticize current approaches from a flow standpoint. Lastly, we close those sections with a *Section Summary*. In this closure, we specify the main challenges and the most promising directions.

Moreover, papers included in those surveys were found by searching the next terms:

> (“Immersion” OR “flow”) AND (“personalization” OR “player experience modeling” OR “experience-driven” OR “User modeling” OR “player satisfaction” OR “real-time adaptation” OR “player behavior” OR “dynamic difficulty scaling” OR “procedural” OR “generation” OR “entertainment modeling”)

In the following databases: *ACM*, *IEEE*, *ScienceDirect* and *Google Scholar*

In addition, we included some articles that were cited in the selected papers. To narrow the scope, we focused on empirically tested approaches. Lastly, we give our last statement about flow and game AI in Section 10.

3. The theory of flow

In this section, we present a brief introduction to the Theory of Flow. We focus on explaining its main constructs and how they have been evolving over time. Furthermore, we give examples of contexts in which it has been successfully applied and evaluated.

Back in the mid-seventies, flow research started in a endeavor to understand this subjective state, characterized by an intense level of attention, that people experienced while performing an intrinsically motivated activity [7]. This phenomenon was called flow, and it is defined as follows:

> Being “in flow” is the way that some interviewees described the subjective experience of engaging just-manageable challenges by tackling a series of goals, continuously processing feedback about progress, and adjusting action based on this feedback. – Nakamura and Csikszentmihalyi [17].

Also, it has been found that a subject can experience a state of flow in a variety of activities [4,6,16,17], such as: dancing, rock climbing, playing chess or a video game. In addition, it was observed that a state of flow could be described by the following characteristics [7]:

- Full immersion in the task
- Feeling of full control
- Being less conscious of the passage of time
- Sense of identity lessens, but is reinforced afterwards

However, all these characteristics are not necessarily present in a flow state. The only defining quality of flow is that attention is fully invested in the task at hand. Also, these investigations have discovered that flow is a state of optimal experience; people enjoy the most when achieving the subjective state of consciousness known as flow [7]. Arguably, this consequence of achieving a flow is the main cause of the motivation to adapt flow constructs to a video game context.

Further research identified that when a subject performs a low skill and challenging task, apathy is experienced although she is in the flow channel, according to the original model of flow [7] (Fig. 1). Because of this finding, the conditions needed to drive a subject to a flow state changed, and the eight channel model of flow has been proposed [17] (Fig. 2). According to this new model, a subject may be able to achieve a flow state if the perceived challenge and skills are above his/her mean levels [17]. In Fig. 2 it is shown that when challenge and skills are below the subject’s mean, apathy is experienced.

Another aspect that has been identified as key to achieve a flow state is an autotelic personality of the subject. Although the skill to achieve a state of flow is present in almost all people, the frequency and quality of the experience vary from subject to subject [17]. It

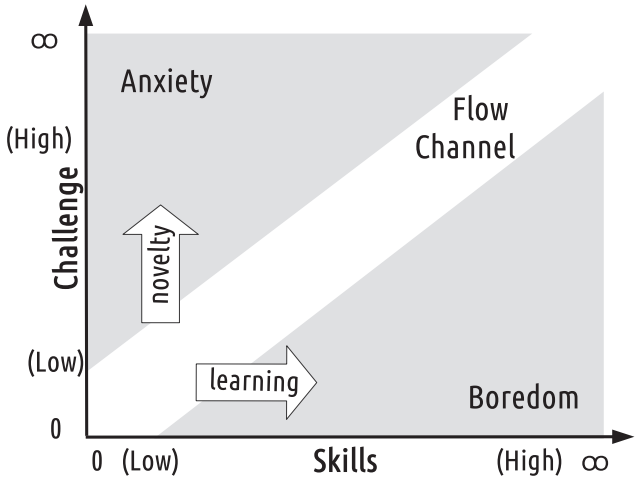


Fig. 1. The original three-dimensional model of flow (reproduced from [7]). In this figure, we can see how important is the challenge/skill balance precursor. If the player does not stay in the flow channel, his/her experience is not optimal. If the challenge she is facing is *below* his/her skills, *boredom* is experienced. Conversely, if the challenge she is facing is *above* her skills, *anxiety* is experienced.

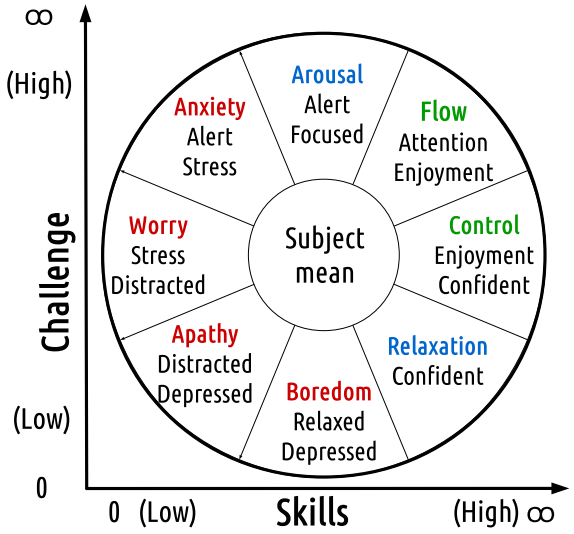


Fig. 2. The eight channel model of flow (reproduced from [17]). This model is less coarse than the original model of flow, which leads to a better description of the emotional state of a user according to the challenge/skill balance she is experiencing. Also, unlike the original model, when challenge and skills are below the subject's mean, apathy is experienced.

Table 1
The eight elements of flow. These are empirically found factors that foster flow (precursors), and certain player traits are essential to experience the subjective state of mind called flow.

Flow elements
A challenging but tractable task to be completed
One is fully immersed in the task, no other concerns intrude
One feels fully in control
One has complete freedom to concentrate on the task
The task has clear unambiguous goals
One receives immediate feedback on actions
One becomes less conscious of the passage of time
Sense of identity lessens, but is afterward reinforced

has been observed that the ones that experience flow more frequently, and stay longer in it, have an autotelic personality. This kind of personality enables the subject to be motivated to perform a task by its own will.

We already explained that flow is a subjective experience that most people can achieve. Another fundamental component to achieve flow is player traits; it is desirable that players have an autotelic personality. Also, we describe one core precursor of flow: the challenge/skill balance. In addition to this precursor, there are eight more constructs of flow that any activity, intended to facilitate an user to get into the flow, should include. In Table 1 we show these elements, originally proposed in [7]. Thus, if we design an activity to include all these heuristics, we would break all the barriers that could block a person to experience flow. However, depending on the subject, not all of them are necessary to clear the path to flow.

Lastly, flow is a state of optimal experience. Its precursors are heuristics, which facilitate the subject the achieving that state. It may seem easy to translate these heuristics to a video game context, however, it is a hard to do it at an elementary level.

We have introduced the basic concepts to understand flow, its elements and models. Also, we have briefly explained the outcomes of achieving a flow state and how it can impact player experience in a positive manner. In the next section, we broadly describe the layout of this article and the methodology conducted in order to select works for our survey.

4. Adapting the theory of flow to a video game context

An effective adaptation of the eight elements of flow to a video game context would bring great advantages to the player-centered game research area, because of the consequences (outcomes) of achieving a flow state. The main consequences of a flow state are: the subject's attention is fully invested in the task at hand [7] and enjoyment [17]. Depending on the context, we can find even more desirable outcomes. For instance, in an e-learning system context, it has been found that a state of flow has a positive impact on learning rates [16,19,20]. Therefore, a clear understanding on how to use the flow constructs in a video game context would enable us to effectively shape all kind of video game components (space, mission, NPCs, etc.) so they facilitate the player's achievement of a flow state.

Over the years, there have been several attempts to adapt The Theory of Flow to the context of video games [4,6,12–14,16]. We can categorize these approaches in the next groups:

- Linking antecedents of flow to game design methodologies [4].
- Criteria to objectively evaluate the capabilities of a video game to foster flow [13,14].
- Mapping antecedents of flow to gameplay attributes [6,12–14,16].
- Gameplay experience models based on flow constructs [6,16].

4.1. Linking the antecedents of flow to game design methodologies

The work in [4] is the only one, to our knowledge, that follows the methodology of *linking the antecedents of flow to game design methodologies* (Table 2). Although this approach is not suitable for the purpose of our work, it is worth mentioning its results and evaluation methodology.

In order to test the validity of the proposed game design methodologies, they designed the video game FLOW™[1] around them and released it online with great success. More than 350,000 copies were downloaded within the first two weeks after the release. Then, they read the comments from players looking for key words describing a flow state.

Although this methodology is informal and purely qualitative, they got outstandingly positive results. They reported that "addicting" was the most common word that players used to describe the game. Also "relaxing" was a recurrent word. This could mean that this game fosters flow for the particular player type that enjoys the most (or get into flow) while feeling relaxation.

4.2. Criteria to objectively evaluate the capabilities of a video game to foster flow

The authors in [13] proposed the GameFlow model criteria, which is used to objectively assess how good a game is at fostering players to achieve a flow state. GameFlow is based on the next eight elements: concentration, challenge, skills, control, clear goals, feedback, immersion, and social interaction. Also, it has been designed to be general or applicable to all game genres, although it was only evaluated by using real-time strategy games.

Some years later, a specialized GameFlow model was proposed [14]. In this iteration, the criteria was specified to evaluate real-time strategy games. This specialized version of GameFlow was proposed because the first one demonstrated to be inadequate for a general evaluation method.

We found two main problems with the last approach. First, a purely objective evaluation is inappropriate to assess the subjec-

[1] http://interactive.usc.edu/projects/cloud/flowing/.

Table 2
Game design methodologies, proposed by [4], based on flow heuristics.

Game design methodologies to achieve flow
Expand your game's flow coverage by including a wide gameplay spectrum with different difficulties and flavors
Create a Player-oriented Active Dynamic Difficulty Adjustment (DDA) system that allows different players to play at their own paces
Embed DDA choices into the core gameplay mechanics and let the players make their choices through playing

tive experience that is flow. Since each type of player experiences flow, or enjoys a video game the most, in a different manner. Second, they consider that single-player games should be penalized because they lack of the social interaction element. Although this asseveration may sound accurate, because social interaction is an implicit element of flow [7,6], it has been improperly translated to the video game context. Single-player games satisfy this element of flow trough social interaction, outside gameplay experience, that may contribute to the intrinsic motivation of a player to play a particular game [21]. Additionally, there are many experiments that confirm that single-player games are effective at fostering flow [16,22,23]. Therefore, we argue that a multiplayer experience is not an essential component for a game aimed to facilitate the player's achievement of a flow state.

4.3. Mapping antecedents of flow to attributes of gameplay

Mapping antecedents of flow to attributes of gameplay is the most common approach. In fact, it is arguably the starting point for all other approaches. Before proposing game design methodologies or a criteria to evaluate games, researchers need to translate the core elements of flow to tangible elements of video games. Consequently, it is vital to clearly understand how to correctly adapt the Theory of Flow constructs to a video game context.

The initial work that followed this method can be found in [12], which is the continuation of [24]. The need for more engaging Computer-based Learning Environments (CBLEs), for all kind of students, is the reason for doing these research works [12,24]. In the first article, chronologically speaking, the author conducted an empirical investigation in order to find out what game features promote engagement on players. His hypothesis was that knowledge on how video games engage players could be extrapolated and be used in CBLE environments. The results of this study drove the author to relate the similarity between constructs from the Theory of Flow and the game design methodologies behind video games that "many people enjoy and play for hours".

Then, in [12] a mapping between the eight elements of flow and gameplay features is presented, however, its analysis is limited [6]. Both articles [6,16] present a formal mapping from the original flow elements to gameplay features. Also, they describe the interactions between players and a flow-based video game. Consequently, we only include these two efforts in the textual comparison shown in Table 3. In addition, we present a mapping of the eight elements of flow that condenses these two approaches and some ideas from the original Theory of Flow:

1. *A task to accomplish:* The adaptation of a task element to be accomplished is straightforward, "a task to be accomplished represents the whole gaming experience" [6]. Though, we should consider that depending on the player's typology, for a certain game state, the resulting subgoals, and their priority, will be different [16]. This element reinforces the importance of a subjective approach to assess the level of difficulty perceived by the user. Thus, methods of dynamic difficulty scaling (DDS) that optimize according to player performance, do not necessarily foster players in achieving a state of flow.
2. *Ability to Concentrate on a Task:* We must consider two aspects for this element. First, if a player lacks an autotelic personality she won't be able to get into flow [17]. Second, the interface and controls should be clear enough to foster the player's concentration on the game.
3. *Sense of Control Over Actions:* Players feel control over their actions at different levels. The lower level occurs when players are familiar with the controls and genre conventions [6]. The second level is related to the sense of control of players in the game [16], which arises when players are free to take choices and their effects have a salient impact on the game environment [4].

Table 3
The eight elements of flow and their corresponding gameplay elements.

Flow elements	Cowley et al. [6]	Pavlas [16]
(1) A task to be accomplished	The whole gaming experience (including social interaction during game-play)	The game and the player's perception thereof
(2) The Ability to concentrate on a task	Telepresence [25] and an environment dedicated to gaming	A task requiring concentration as well as an interface that does not impede it
(3) Sense of control over actions	Familiarity/skill with the controller, genre conventions, game-play mechanics	A sense of control is created when the player has a feeling of agency. For this to occur, the player's actions must have a salient effect upon the game's environment
(4) Deep but effortless involvement	High motivation to play, non-imperative to other concerns intrude do otherwise; empathetic to content	It is a result of unimpeded interaction with a skill-challenged tuned system that provides interesting choices
(5) Clear task goals	Missions, plot lines, levels; any explicit outcome of a successful play session	Make goals clear from an interaction-accordance standpoint or some sort of goal listing (e.g., the "quest log")
(6) Immediate feedback	Well-timed, suitable rewards and penalties	Effects of player performance should be evident (player's actions are transparent to the user)
(7) Being less conscious of the passage of time	Focusing on another, temporally-independent environment	The authors of this work argue these two elements describe immersion as an outcome of flow. This state describes a player's state of mind which is characterized by loss of concern for the self, an altered sense of time and a deep but effortless involvement
(8) The sense of identity lessens, but is reinforced afterwards	Embodiment in game avatar; sense of achievement after play e.g."Hi-Score"	

4. *Deep But Effortless Environment:* This flow requirement is related to immersion. A deep but effortless environment requires to remove all barriers that prevent players from immersing in the game. Additionally, in order to accomplish it, we need to offer players the perfect level for their skills and sense of control over actions [16]. We describe the relationship between the concepts of immersion and flow in more detail in Section 9.
5. *Clear Task Goals:* The need for concrete goals has been identified in both the game design and game AI communities, [16,26]. A flow state cannot be achieved or maintained when players do not know what their goals are. Goal theory emphasizes on the importance of clear goals too [27]. The affective state of the players is positively affected when his/her goal is accomplished. Contrary, all obstacles faced in order to achieve that objective result in negative emotions (as cited in [28]). An adaptive game component of missions should be able to generate quests with clear goals for players. Also, NPC companions should be able to identify when a player does not know which is the current goal and assist him/her. If the genre allows it, game mechanics should be tailored to empower the player to choose the goal she wants to pursue. Further, missions or quests should not be imposed to players that are already in a flow state.
6. *Immediate Feedback:* This construct explains the need for well-timed feedback on the player's actions [6,16]. According to [29], the appropriate time interval to initiate a reaction goes from midway through the completion of an action to a maximum of 400 ms after the completion thereof. Furthermore, this flow construct is related to the last element, because players must receive rewards and penalties, if they achieve or fail a goal, respectively.
7. *Being Less Conscious of the Passage of Time:* This element is considered by [6,16] as a side effect of a flow state. Indeed, they define it as part of the emotional experience that is immersion. This experience occurs when working memories of players are completely filled by the game data, thereby altering their perception of time.
8. *Sense of Identity Lessens, But Is Reinforced Afterwards:* This can be considered also as a flow output and immersion characteristic [6,16]. More specifically, this is experienced when the player's personification in the game (avatar) feels like his/herself. Then the sense of identity is reinforced afterwards, when the player has the impression of achievement.

Additionally, *Challenge/Skill Balance* is a core antecedent of flow. This characteristic is key in making flow a deeper and lasting experience. Although this is not part of the eight elements of flow, it is arguably the most important component of an experience that is aimed to foster a flow state [17]. For most players, it is easier to get into flow if they face challenges above their average skill level, but some of them enjoy the most while feeling arousal or relaxation [17] (Fig. 2). Thus, a flow based DDS system should adapt the difficulty level according to the typology of the player (for a survey about this subject, refer to [30]). In general, more avid players enjoy the most when experiencing arousal, while casual players prefer to feel relaxation [4].

We have explained the constructs of the Theory of Flow and the different approaches to translate them to a video game context. In the following subsections we describe, in detail, the model of flow proposed in [16] and the system of flow proposed in [6]. Both methods complement each other. The first is focused on how the elements of flow concepts are related. This facilitates the understanding and visualization of how we should design an effective player-centered game based on flow theory. The second gives us a better insight on player experience and how we should adapt game entities to foster flow. Additionally, this approach [6] provides guidance on how we should model players and propose a standardized way to describe the structure of a game.

4.4. Gameplay experience model: the theory of flow as a model to improve player-centered experiences

Several models of flow have been proposed in different research areas [7,17,31–34] and all of them have in common that the only explicitly analyzed precursor, and its relation with a flow state, is the challenge-skill balance element. However, the effort in [16] excels because, by design, is more flexible and explicitly studies the relationship between all of the precursors of flow and their respective outputs (Fig. 3).

The flexibility of this model facilitates further studies aimed to a better understanding of the relationship between flow constructs in different contexts and applications. Another aspect that makes this work relevant is the fact that it has been empirically evaluated using a video game (more specifically, a serious game). The positive results of the conducted player experience studies and their strong theoretical basis make them adequate as a starting point for further research related to flow and play in games.

The elements of flow that we chose to be empirically tested are: playfulness, video game self-efficacy, challenge-skill balance, emotional experience and flow state. This study is aimed for serious games. Therefore, they also included the constructs of performance, play, and learning; although they are not part of the original flow theory. The majority of the results were consistent with the previously proposed relationships of concept, which were purely based on theory (Fig. 3), yet there was an interesting outcome. The proposed hypothesis says that the challenge-skill pairing element fosters flow, but this was only partially supported. This outcome could have been caused by the objective manner in which the performance of the players was assessed [16]. Also, this reinforces that an objective assessment is not suitable to quantify flow.

4.5. Gameplay experience model: the theory of flow as a system to improve player-centered experiences

This article [6] makes an effort defining the theory of flow constructs as an information system framework. In order to achieve that goal, they begin by identifying the key segments of flow that we should understand in order to adapt this theory to a video game context. These segments are: player interaction, player modeling and game structure.

In regard to *player interaction*, they refer to the emotions players feel while playing a video game (see Fig. 2). These abstract states of mind are explained though lower level emotions. Flow, for example, is characterized by attention and enjoyment. Thus, an effective dynamic difficulty scaling system, designed around the flow constructs, should be able to adjust the difficulty level above the player's mean in order to avoid negative states of mind, such as: boredom, apathy or worry.

As we have explained in Section 4, not every person is able to get into flow while experiencing the same emotions [4,17,35]. Such marked differences in player preferences make inexorable the inclusion of a user model [6]. In essence, a *player model* keeps track of the player's reactions to important gameplay situations. From a flow standpoint, a player model should be designed to identify the emotional experience of users, according to the eight dimensions of flow, in real-time [6,9,18]. Besides, this should be designed to be non-intrusive for users in order to prevent immersion from being blocked.

In order to try to formalize the study of game AI, we need to talk the same language when describing a *game structure*. In [6] proposes a format based on the game's verbs and nouns. However,

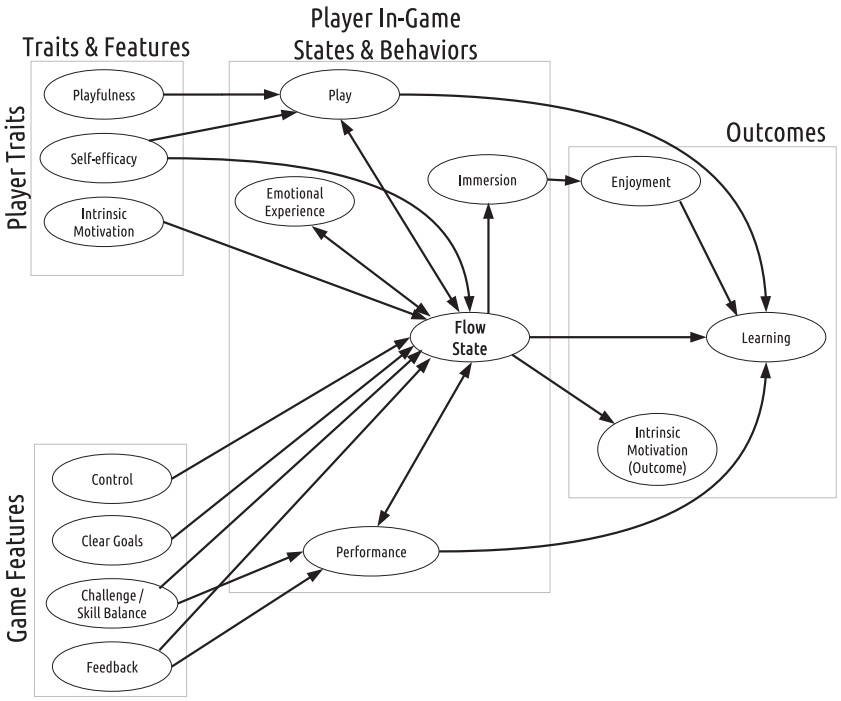


Fig. 3. The theory of flow elements as a model for serious games (reproduced from [16]).

[36] presents the most accepted language in this area to describe a game. Specifically, [36] recommend to construct a collision matrix. All possible interactions between all the characters in the game are described in this matrix.

Lastly, we would like to reaffirm that authors of [6] consider immersion as an output of flow. However, it is still an open discussion whether immersion is a consequence or precursor of flow. It is essential to clarify how we approach immersion since this is a big influence factor to the structure of our proposed flow framework for game AI (Section 6). Therefore, in the next section we argue how we will address the concepts of immersion and flow.

5. Immersion and flow

The similarity between the concepts of flow and immersion has motivated research aimed to a better understand these two emotional states and how are they related to each other [9–11]. The term immersion is the most widely used of the two. In literature and game industry, immersion is considered as the outcome of good gaming experience [8]. However immersion is still a nebulous concept.

In the context of video games, this has been the cause to conduct qualitative studies for a better understanding of what immersion is. Specifically, It is unclear if all players refer to the same emotional state when they use this term, what is causing it, and which are the side effects of reaching this state [8,37,38].

One of the first efforts focused on defining the concept of immersion from the player's perspective is found in [38]. The results of their qualitative study are that players use the term immersion to describe the level of involvement with a game. Also, this empirical study led to establish tree sequential stages of immersion. The first stage is *engagement*, although this concept describes the lowest level of involvement, it is necessary to achieve a deeper level of concentration. The second stage is *engrossment* and the last one is *total immersion*. The last concept defines a state of mind almost identical to a flow state, but different in essence. A total immersion state defines a fleeting experience while flow a lasting one.

The main objective of the article found in [37] is the design of an immersion-based gameplay experience model. The model is focused on identifying the key elements of gameplay that describe the emotional experience of players and also, how can the quality of this experience be assessed. To achieve their goal, they interviewed children that play video games with regularity and their parents, who are not avid players, about the concept of immersion and which gameplay elements contributed to get completely involved in the game.

One of the findings of the analysis of these interviews is that " [...] immersion is a many-faceted phenomenon with different aspects that can appear and be emphasized differently in the individual cases of different games and players". Based on that premise, they designed the SCI-model, which is established on the next three components: sensory, challenge-based and imaginative immersion.

Sensory immersion is related to the audiovisual presentation of video games; a good graphic presentation and sound contribute to get players immersed in the game. The second form of immersion, *challenge-based*, is maximized when players face the perfect level of difficulty for their skills. The last component, *imaginative* immersion, emerges when players empathize with the characters and story of the game. Furthermore, we should consider that the three forms of immersion overlap with each other. For instance, a gameplay element can contribute to all forms of immersion.

In regard to the work in [8], the process followed to design tools intended to assess – in an objective and subjective manner – the level of immersion of players in a game, drove the authors to define immersion as "an experience in one moment in time and graded (i.e. engagement, engrossment, total immersion). Immersion involves a lack of awareness of time, a loss of awareness of the real world, involvement and a sense of being in the task environment".

This definition is based on concepts of flow and cognitive absorption. Also, they claim that flow and immersion are very similar concepts yet different. The first difference that they point out is

that flow is an extreme state of mind and immersion is not always so deep. Therefore, it can be considered that a state of flow is the deepest level of immersion (total immersion in terms of [38]). Authors refute the equality of these terms arguing that there are games without some key elements of flow which are considered to be very immersive.

However, according to flow theory, the last argument – which refutes that total immersion and flow are different concepts – is incorrect. Flow is a subjective state of mind that can be achieved by almost any person. For a subject with an autotelic personality is specially easy to get into flow. The eight elements of flow are usually present in activities that foster flow but not all of them are necessary and are not part of the definition of a flow state. These elements can be considered as heuristics to design activities aimed to facilitate users to achieve flow.

Likewise, it doesn't imply that flow and total immersion concepts describe the same emotional state. Although their definitions are basically identical, there is a key difference between them: total immersion is a fleeting experience while flow is lasting [38]. Indeed, total immersion is more similar to micro-flow [37,39]. Consequently, in a game AI context, we can extend the duration of micro-flow, or total immersion, through an effective dynamic difficulty scaling module.

The work in [9] supports what we propose in the previous paragraph. In that article, they define flow as an optimal experience that occurs when one feels immersion and engagement at the same time. They argue that immersion emerge from recognition of familiar schemas, and engagement from interaction or decision making. Moreover, they claim that in order to keep the user engaged, the difficulty of the schema should be gradually incremented. Because engagement is lost when a subject fully understands a schema [9].

Therefore, for the purposes of this document, we consider immersion as a precursor of flow. More precisely, we focus on game features that facilitate immersion through gameplay and AI techniques. Immersion can also be enhanced by audiovisual elements. However, that kind of components are not suitable for flow constructs, since they do not affect the experience of playing a game. This perspective of flow and immersion, and the previous review of works aimed to translate the flow constructs to a video game context, enables us to propose a flow framework for game AI.

6. FlowAI: A generic flow framework for game AI

Analyses from previous sections have empowered us with a better understanding of flow in a video game context. We consolidate this knowledge on our flow framework for game AI (FlowAI). Our framework unifies the key elements of preexisting approaches that better suits a game AI context.

We structured the FlowAI framework as a generic player-centered game. Therefore, its elements are condensed in three modules: player modeling, dynamic difficulty scaling (DDS) and immersion (see Fig. 4). Further, we detail the relation between these modules in a functional manner.

The *player model* module takes as input the data necessary to describe the player's typology and his/her emotional experience while playing the game. As output, this module drives, how and when, the DDS and/or immersion have to tailor its corresponding gameplay elements. Hence, a player model is essential for any effective player-centered game.

Regarding the *DDS* module, it is responsible of adapting any tangible gameplay feature that might change the perceived level of difficulty. Thus, we have proposed that this module is able to adjust any combination of the following gameplay features: NPC behavior, quests or scenarios and game mechanics. Unlike other flow frameworks, ours considers immersion as a precursor of flow. The *immersion* module focus on breaking all the barriers that could block a player to immerse in a game by adapting: NPC behaviors, quest or scenarios and controls.

Despite the fact the DDS and immersion modules might adapt the same game mechanics, they achieve their goal in a different manner. These modules are not mutually exclusive. For instance, let us consider we have a player-centered game that implements a player model, DDS and immersion modules; all working together to personalize the NPCs behaviors. While the DDS module would be balancing the level of difficulty, the immersion module would prevent the NPCs from performing in an unbelievable manner. Each module, adapting a specific game mechanic, breaks a different barrier that might block a player from getting into flow.

Another feature of the FlowAI framework that we would like to emphasize is that each of its modules can be interpreted as game AI subareas. This level of abstraction lets us explain the directions of game AI techniques intended to design better flow-based games. Therefore, through the lenses of our framework, we proceeded to survey: player modeling, DDS and immersion subareas in Sections 7–9, respectively.

7. Existing work on player modeling

In this section we survey current trends on player modeling. Our evaluation method for this analysis is based on the FlowAI

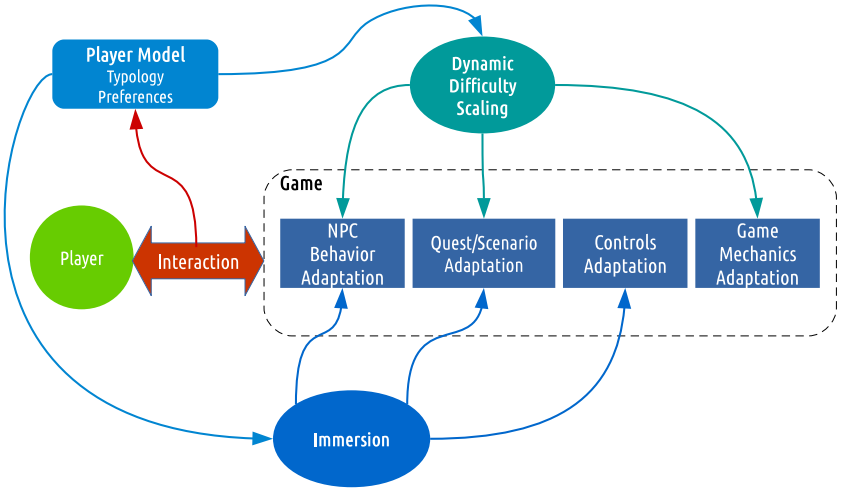


Fig. 4. FlowAI: a generic flow framework for game AI.

framework (refer to Section 6). In particular, we have the main objective of identifying the most promising methods to design an effective *player model module*. We remind the reader that the selection criteria of articles is described in Section 2.

The player modeling area is so extensive and diverse that we use the survey [40] as a starting point. Namely, we use their categorization methodology for player modeling approaches since it fits ours. Also, in this manner, we can use a the same language of this – remarkable and up-to-date – survey about techniques aimed to personalize games.

Karpinskyj et al. [40] propose to classify player modeling approaches according to both the data used as input and the gameplay features that are being tailored. For the propose of this section, we are only interested in the classification from the player data standpoint. This is the proposed categorization:

- *Player preferences* – player preference classification is made according to the player's taste in video games. More specifically, it typifies players based on their personality or preferences for certain video game characteristics.
- *Player personality* – category is very similar to the one above, since it also typifies players in accordance to their personality. The difference is that in this case, psychological personality models are used, such as the Five-Factor Model [41] and the Myers-Briggs Type Indicator [42]. For adaptive games based on preference player data, new personality models are proposed specifically in order to describe different types of players.
- *Player experience* – category groups all the work using player data to infer their emotional experience while playing a game. The inference of player experience can be done by correlating the game player's actions and data from physiological measures or self-assessment reports.
- *Player performance* – this type of model keeps track of how well a player is doing in a game. That is, these works focus on describing players according to their performance in the game.
- *In-game player behavior* – This category is very similar to the player experience one. However, in this case the player experience is deduced using only the player's in-game behavior. There is no other data input to correlate to.

Now, let's consider we want to design a player model – that fits the FlowAI framework – with the objective of driving DDS and immersion modules. In order to be capable on fostering any type of player into flow, this player model should combine many types of approaches from the previously explained categorization.

First, we need to understand the *player preferences* [43–45] towards the game features we intend to adapt. This characterization needs to get data from players without breaking immersion. Regarding *player performance* [35,46–48], we must know how players perceive the challenge of a particular game. We would like to emphasize that it is required in order to interpret the player's perception of difficulty; an objective measure of performance is improper for this context.

The player preferences that we just described can be grouped as player typification features. Therefore, this aspect of the player model may be achieved entirely offline. That is, this stage may be completely accomplished before the player starts playing the game. However, the player's typification could be refined using in-game data, like the approaches in the *player experience* or *in-game player behavior* categories.

Also, in order to prevent a short-lived flow experience, a real-time assessment of the player's emotional experience is a must [6,9,18]. Since it is essential to recognize the player's emotional state according to the challenge/skill balance she is experiencing (see Fig. 2). This feature is related to the *player experience* category.

A player model designed around these features should be reliable at facilitating the player's achievement of flow state; regardless of their age, skill level or gender. From this standpoint, we proceed to review the selected works on the player modeling research area. That is, we focus on approaches that follow a methodology that could be used for flow-based games. Further, we discuss the most promising methods that would endow us with all the desirable features we have described.

In the first place, how do we get preference data from players in an accurate and reliable manner without breaking immersion? According to [49], the most commonly used techniques to get preference data from players are: subjective self-reports and physiological measurements. Both approaches have proven to be accurate in most cases [40,49] but some methods do not maintain immersion. To prevent this from happening, we should avoid using intrusive hardware for physiological measures and applying questionnaires in between gaming sessions (for an overview refer to [49]).

Work with the aim of acquiring more accurate player data without breaking immersion has been stable in recent years [2,44,50–55]. Of these approaches, the most significant, from our flow standpoint, is [2]. The main goal of [2] – which is vaguely based on flow theory – is to implement a Procedural Content Generation (PCG) system that personalize stages for *Infinite Mario*. The main objective of this work is to maximize one particular emotion in players. The emotions selected to be maximized are; engagement, frustration and challenge.

The player model is created by fusing visual and behavioral cues [2]. The visual cues that were used are: mean head movement and visual reaction features. For behavior cues, relevant gameplay events were used, such as; interaction with game items and level completion. Player preferences are collected via subjective self-reports and the player model is constructed using a neuroevolution approach (a genetic algorithm that evolves an artificial neural network).

Although this article [2] is in an incomplete state – since it has not been properly validated in addition to the lab setting bias – it is the player-centered application that incorporates more flow elements to date.

Another alternative that has been little explored in academia is *self-typification*. This approach consists on directly asking the players for the preference data we need. Most of the commercial games, of the last three decades, have implemented self-typification. This was done by directly asking the players, before they start playing the game, the level of difficulty they want to face. Although most of the commercial implementations are imprecise because the broad classes they use (easy, normal and hard), their commercial success suggests that this simple player classification method makes games appealing to a wider range of players. Besides, there is empirical evidence in academia that encourages the use of self-typification methods [4].

For instance, the hybrid approach in [47] includes preference and in-game behavior player data to generate levels for an action game. The preference data focus on the preferred level of difficulty (easy or difficult) and preferred weapon type (short or long range). Then, players are typified by a k-means algorithm with a $k = 2$ for best results. The second stage of their approach, consist on constructing two näive Bayes classifiers that use player type as prior knowledge and player in-game behavior as input. One classifier is built to identify frustrating levels and the other for boring ones. In order to avoid noise on the players responses a ranking algorithm was implemented to manually assess generated levels.

These models were constructed using 80% of the dataset for training and the rest for validation. The frustration and boredom models achieve an accuracy of 91.6% and 87.5% respectively. Also,

they conducted an experiment in which they generated missions intended to foster flow. These missions were generated using the solutions that minimized both boredom and frustration. To determine if these flow scenes maximize enjoyment, players were asked to compare them against other randomly generated game levels. 75% of testers enjoyed more the flow version of the game levels. Because of how the test was conducted, the results do not indicate that most of the players achieved flow, they just suggest that they preferred a medium level of difficulty.

They [47] also attempt to find what preference data and performance measurement characterizes the players in both categories found by the k means algorithm. It is interesting that most of the players classified in group 1 performed significantly better than those in group 2. Furthermore, 70.7% of group 1 players replied that they preferred to play difficult levels and 60% of group 2 players replied they preferred easy levels. These results may suggest that there is a relationship between the preferred level of difficulty and the objective performance of players.

A very unique approach is presented in [35], which adapts the level of difficulty of a turn-based role playing game. They propose a tensor factorization model that constructs a long-term set of challenges based on player's performance. Although the adaptation mechanism is tailored according to the objective player's performance instead of the subjective perceived difficulty, this method could be used for an application based on flow heuristics. The reason for this is that – at least for the tested game and the demographics of the players that participated on the tests – they proved that there is an inverse relationship between the perceived level of difficulty and the objective performance of players.

7.1. Section conclusions

The starting point for this analysis is the player model we defined, which fits our generic flow framework for game AI (Section 6). Thereafter, we identified that the most adequate techniques to get data from the player, without breaking immersion, are: self-typification, subjective self-reports and physiological measures using non-intrusive hardware. Specially, self-typification approaches have been hardly explored, even though they are promising.

Also, we found that a fundamental trait a player model should store is the player's learning rate. This feature is essential to estimate the optimal frequency to adapt game features; which depends on the game genre and type of player. If the optimal frequency is met, it means that the player's learning rate matches the increase rate of difficulty.

Another aspect we should consider is the main goal of the video game we intend to design thorough flow constructs. For instance, for purely ludic games, a subjective assessment of difficulty should be the right way to go, but for serious games this might not be the case. According to [56,57] – in a context of serious games for rehabilitation – it is better to adapt the difficulty level of games using objective measures to maximize the recovery of patients. However, [58] states that enjoyment is needed to get better learning results in the long-term. Hence, [59] proposes a hybrid method that combines the user-guided and task-guided mods to ensure effective short-term and long-term learning results in serious games.

We have noted a generalized use of lab settings in evaluation methods. This may bias the results. Players should be able to play the game with their preferred setup and time.

8. Existing work on dynamic difficulty scaling techniques

Dynamic Difficulty Scaling (DDS) modules can be used to tailor any gameplay feature that contributes to the challenge in a game. We can broadly categorize DDS modules, accordingly to the gameplay features they adapt, in the next three groups: NPC behavior, quest/scenario or game mechanics. Based on this categorization, we surveyed existing work on dynamic difficulty scaling techniques that directly affect skill-based challenges. In particular, we review DDS methods through the lens of the FlowAI framework (Section 6). Also, we put special attention on what player data is used as input, the adapted gameplay features and the techniques used to perform this adaptation. Also, we point out the most promising directions that will endow us with a better understanding on how to design an effective *DDS module* for the FlowAI framework. We remind the reader that the selection criteria of articles for this section is described in Section 2.

8.1. Quest/scenario

The hypothesis tested in [48] is that their hybrid approach – that consists on a constraint solver and evolutionary system – is capable of automatically generating game levels for Super Mario Bros.™ that match the quality of those created by professional game designers. The metric used to measure the quality of the game levels that were designed, was their rhythm-group model, which is based on [60]. The general idea of this model is to challenge players with oscillating difficulty levels throughout the whole game level. The values of three variables that determine the frequency of the difficulty and the maximum peak value in a defined period of time are set by hand. In order to create levels similar to those found in commercial games, they used the values found by a supervised machine learning algorithm trained to differentiate between well and bad designed levels. Also, this classifier was used to evaluate the quality of level design generated by their system. The results – according to the classifier and the user tests performed during the 2010 Mario AI Championship Level Generation Track competence – were positive.

The application proposed in [48] is intended to work offline and it needs previously designed levels, however, it can be adapted in a way that better suits the flow heuristics. The mayor adjustment would be to change the metric used to measure the quality of the game. The objective of this metric would be to keep a straight difficulty curve [35]. Therefore, the difficulty of the level would increase at the same learning rate of the player.

Smith et al. [61] proposes an offline DDS system that tailors the position and orientation of obstacles in the stage and NPC behaviors, with the objective of minimizing the difference between the expected and desired level of difficulty specified by the author. The predicted level of difficulty is measured by statically analyzing the expected interactions between the NPC and the player. Rather than using real data to assess the level of difficulty along the path followed by the player – from the spawn position to his/her goal – they automated the process by replacing the player by a bot. This NPC representing the player is endowed with a simple behavior to determine the most likely path a player would take to complete the quest. Therefore, the input for this DDS module is only in-game player data. In regard to the chosen optimization method, they reported that the Covariance Matrix Adaptation Evolution Strategy technique (CMA-ES) is particularly useful in this domain because the problem is ill-conditioned and noisy. Although, this application is a tool aimed to assist game designers in the laborious task of testing the game, the good results they have achieved suggest that the CMA-technique can be successfully used for player-centered games.

In [62] we can find a similar procedural content generation system that creates game levels based on difficulty curves. They use a genetic algorithm to optimize the difference between the desired difficulty level (assigned by the game developer) and the objective difficulty level measured by a metric. This metric was constructed by experts.

It is worth mentioning that the results from the hybrid approach in [47] (already described in Section 7 and [35,63] suggest that there is a relationship between the preferred level of difficulty and the objective performance of players. Furthermore, a shortcoming of article [47] is that they do not consider the subjectivity of the emotional state of flow, despite the fact that they already have a typification system for players. It may explain a lower accuracy level compared to the other versions of the DDS system.

8.2. NPC behaviors

Spronck et al. [64] proposes an online stochastic optimization method intended to improve scripts that generate NPC behaviors. This approach, called Dynamic Scripting, is similar to the reinforcement learning algorithm. However, unlike this, their method works well in environments that do not satisfy the Markov property, such as the role-playing game (RPG) that they used as a testbed. Also, they define three different strategies of how the weights of the strategies are updated in order to adapt the level of difficulty to match player's skills (based on her performance). They named these strategies as: high-fitness weight penalizing, weight clipping and top culling. The goal of this DDS method is to achieve a 0.5 winning ratio against varied static game AIs with unique challenges and strategies. The top culling strategy outperformed the others by accomplishing means within the range of 0.45–0.55 against all of the static game AIs. The top culling strategy employs an adaptation mechanism that consists on ignoring rules with weights beyond a predefined value ($W_{(}max$)). Weights are updated every time a match ends, if the player wins, $W_{(}max$) is increased by a 10% and vice versa. Thus, the most effective rules are discarded when the Dynamic Scripting algorithm creates strategies for the next match.

The Dynamic Scripting technique has proved to be effective in different video game genres. For example, the winner bot of the first place in the Fighting Game AI Competition 2014[2] is based on dynamic scripting. Also, this method has the advantage of rapidly adapting, therefore, it is suitable for online adaptation. One drawback of this approach is the lack of diversity in the produced behaviors, which may break immersion.

Olesen et al. [65] uses neuroevolution of augmenting topologies (NEAT) to generate NPC behaviors – for the real-time strategy (RTS) game *Globulation 2* – that can adapt its level of skill in real-time. The aim of the adaptivity mechanism is to minimize the difference between the level of challenge of the NPC and its opponent. The level of challenge is computed using a quantitative metric proposed by experts on RTS games. The first conducted experiment to validate their proposed DDS method consisted on facing the NEAT based NPC against a rule-based AI that the game includes. For this test, the fitness of the population was evaluated about every 6.6 min. The results show that the error is kept small only at the end of the evaluation period. Thus, in order to try to improve the results, they proposed to use a NEAT method with a real-time learning mechanism (rtNEAT). The fitness of population for rtNEAT is evaluated about every 12 s. The rtNEAT based DDS mechanism clearly outperformed the NEAT counterpart.

From a theory of flow perspective, the updating rate of a DDS system should be customized for each player. This work [65] gave us a frequency that can be used as a start point for fast paced video games. Furthermore, the authors reported that players did not notice a change in the level of difficulty of NPCs. This problem should arise because the challenge metric was not cross-validated with human perception about challenge in this game.

[2] http://www.ice.ci.ritsumei.ac.jp/ftgaic/index-R.html.

For the sake of completion, we present the work found in [66]. Here, a game AI for a fighting game purely based on anticipation is introduced. In addition, this NPC has the ability to perform an online adaptation of its skill level to match the player's strength. The only input for this DDS module is both performance of the NPC and the player.

This AI follows a two-step process to choose which strategy to take. First, the AI decides if it is about to be attacked, by the opponent using heuristics that are based on statistics of previous attacks in the same game state (same relative position between the two players). Second, if an attack is most probably coming, the AI predicts all the possible game states – and their outcomes within the next second – using a k-nearest neighbor algorithm. The outcome is a value representing the expected global damage, a positive value means that the AI does more damage than it receives.

In regards to the DDS module, it is aimed to keep the NPC's performance just above its opponent. The heuristic to keep this balance uses the NPC performance as input. Therefore, if the NPC score is below a predefined threshold, it will try to outperform its adversary by executing the predicted action that leads to the best benefit. Alternatively, if the NPC's score is above that threshold, it will choose an action with an expected negative damage. The results of the AI performance and its DDS module were both positive. The AI that intended to choose always the best strategy, outperformed the top three AIs in the 2013 Fighting Competition. Since the AI was tuned to perform slightly above its opponent, we can say its objective was achieved.

8.3. Game mechanics

The only work that we found, proposing to adapt the level of difficulty of a game by tailoring a game mechanic is [53]. This work investigates how different camera viewpoints affect the emotional experience of the player. Specifically, the examined emotional states are: fun, challenge, boredom, frustration, excitement, anxiety and relaxation. Indeed, these emotional states seem vaguely based on the theory of flow. In regard to the input, they use objective measures: heart rate, blood volume pulse and skin conductance signals. However, these objective emotional states were not directly used to adapt the level of difficulty, they were correlated to subjective self-reports. They found that the perceived level of difficulty is linked to the average inter-beat amplitude. Also, they empirically found that players find more challenging to play in stages with a camera placed closer to the floor.

8.4. Section conclusions

Table 4 shows all the articles reviewed for this section. These articles are categorized by the gameplay element that is being

Table 4
Reviewed articles on dynamic difficulty scaling personalization, are categorized by the gameplay element that is being tailored, whether this adaptation is performed online or offline, and the type of measures that model the player. Except for [53], because it is in an unfinished state (refer to Section 8.3).

Quest/scenario	Online adaptation based on objective measures [61] Online adaptation based on subjective measures [47] Offline adaptation based on objective measures [59,62] Offline adaptation based on subjective measures [48]
NPC behavior	Online adaptation based on objective measures [61,65,66] Offline adaptation based on objective measures [35,64] Offline adaptation based on subjective measures [35]
Game mechanics	Objective measures correlated to subjective self-reports [53]

tailored, whether this adaptation is performed online or offline, and the type of measures used that model the player.

The dynamic adaptation of difficulty is an optimization problem – driven by a player model and delimited by design constrains – with the aim of minimizing the difference between the desired and perceived level of difficulty.

In the articles we have reviewed, this optimization problem was tackled with evolutionary algorithms [48,61,62], machine learning techniques [47,64,66], heuristics [59,66], NEAT algorithms [65] or tensor factorization techniques [35]. These techniques were chosen in order to cope the specific challenges and requirements found in a game AI context, such as: high dimensionality, rapid online learning, highly constrained problems or missing data. Furthermore, if a player model uses data from both, an analog sensor and in-game behavior, we face a data-level fusion problem with dissimilar signals (for an overview refer to [67]).

Additionally, while theoretical and empirical studies indicate the importance of a subjective assessment of the level of difficulty, we identified that player typification has been barely explored.

Finally, we indicate the features that, according to the FlowAI framework, should considered by a dynamic difficulty scaling module:

- Player typification – Arguable the most common misconception in flow-based DDS approaches is to adapt the difficulty level in the same manner for all kinds of players,although flow is a subjective state of mind. Additionally, the level of difficulty is perceived in a subjective manner. Thus, it is essential to understand both the difficulty preference and how it is perceived by each type of player.
- A quantified notion of challenge – We need to identify what gameplay features contribute to skill-based challenges. [68] proposes a formal methodology and objectively measurable definition of difficulty for challenges that can be personalized for each player. Since the perception of difficulty is subjective, we have to cross-validate these game features with subjective self-reports. That is, we need to understand better how a challenge is perceived by every type of player.
- Feature selection – A feature selection algorithm is useful to identify and dispose of game features that have no significant contribution to challenge. This procedure can lead to reduce the dimensionality of the problem.
- Fast and reliable optimization algorithms – that cope with the aforementioned challenges and requirements. For fast paced video games, an algorithm fast enough to run in real-time is recommended.
- Constant difficulty curves – This means that we prefer that the player always perceives the same level of difficulty, the one that better suits his/her preferences. Therefore, the ideal difficulty curve for a "hardcore" player would be the line labeled as Player C in Fig. 5. The line is straight because the increase rate of difficulty equals the learning rate of the player.

9. Existing work on techniques that foster immersion

In Section 5 we explain our approach towards the concept of immersion. Specifically, we supported our assertion about considering immersion as a precursor of flow. Then, in Section 6 we proposed our flow framework for game AI (FlowAI), which includes an immersion module that is responsible of breaking all the barriers that could prevent a player from immersing in a game by adapting: NPC behaviors, quest/scenarios or controls. Particularly, it focuses on tailoring gameplay aspects that directly affect the experience of playing a game. In terms of [37], this module is aimed to enhance the sensory immersion of games through game AI techniques.

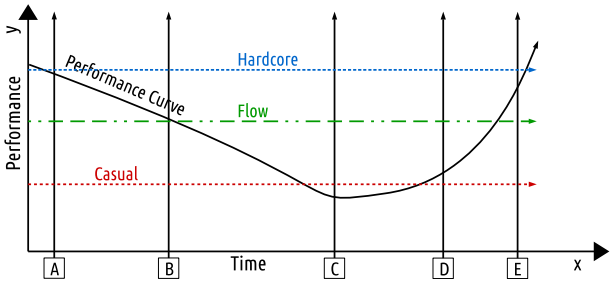


Fig. 5. This figure has been reproduced from [35] and modified to include three more difficulty curves labeled as: Casual, Flow and Hardcore. To explain the meaning of these difficulty curves, we change the *y*-axis to be the perceived level of difficulty instead of the performance. From this perspective, the *Flow* line represents the ideal difficulty curve for a player that enjoys the most while being in a flow state. While *Hardcore* and *Casual* lines represent ideal difficulty curves for experienced and amateur players, respectively.

Sensory immersion is related to the audiovisual execution of games that are improved by realistic and high-quality graphics and sound [37]. Game AI can contribute to create a more realistic environment by generating believable NPC behaviors and/or scenarios. That is to say, NPCs that appear to be controlled by a human, and procedurally created scenarios that match the quality of those created by humans.

The main motivation to create believable content is the strong impact it has on driving players into a deeper level of immersion [69,70] and, ultimately, to a flow state. There is empirical evidence that indicate that players prefer to: play with or against human-like NPCs [71] and play human-like procedural generated content [48]. Besides, we can adapt the game controls to better suit the players' preferences. Personalized controls may prevent blocking players from experiencing a deeper sense of immersion [72].

In this section we review artificial intelligence methods designed to generate believable behaviors or scenarios. Our approach for the analysis is similar to the one followed in Sections 7 and 8. Nevertheless, we relaxed the survey methodology (see Section 2) to narrow the scope; in such a manner that this section is more an introduction to understand what human-likeness is and how this problem is commonly tackled. The approach for this section is valid because an *immersion module* would be effective as long as it generates believable content. However, if the method refines the quality of the content using online data or players feedback, we must ensure that immersion is preserved by using: self-typification, subjective self-reports and physiological measures using non-intrusive hardware.

When we say that an NPC exhibits a human-like or believable behavior, we mean that it looks like it is being controlled by a human. In order to create believable content for a specific game, first we have to understand what makes game AI to appear human-like in that context. Usually, this knowledge is given by an expert or by directly asking players. In [69,73], for instance, they recorded gameplay footage of a first person shooter game with both AI and human controlled NPCs. Then, expert players were asked to identify which bots were controlled by humans and explain which behavior characteristics made them look different against their AI controlled counterparts.

Another method to tackle this problem is using content or behaviors created by humans as input for a supervised machine learning algorithm [48,74,75]. This approach somehow skips the step of identifying what characterizes a believable game AI, since they are based on the hypothesis that a bot/game-level with knowledge – learned by observation – naturally behaves in a human-like manner. On the other hand, the main disadvantage of these methods is that they require several observations of all the possible situations the AI may face.

The second stage consist on interpreting the believable characteristics of low-level features of the AI algorithms intended to generate these human-like behaviors or content. That is, we must find a way to drive AI techniques to generate believable content. For example, if we want to learn believable behaviors using an evolutionary technique, we need to translate the subjective self-reports to a set of rules that restrict the algorithm to explore actions that a human would not perform. In a very broad manner, we can categorize the techniques for learning game AI into three groups; learning from observation [75,74,76], evolutionary algorithms [61,62,77–79], and rule-based methods [80–82].

In recent years, approaches aimed to generate NPC behaviors that are based on cognitive architectures [71,83–85] have gained importance in the community. The main benefit of this approach is that they are actually creating intelligent machines that are prepared to handle unseen problems in a complex and dynamic domain. The main disadvantages of this method are: the complexity involved both in design and implementation [71] and the performance of bots with a large rule set [86].

Finally, we have to evaluate the human-likeness of the generated game AI. Because believability is a subjective concept, the perceived human-likeness of an NPC behavior or a game level depends on both the player and the game being played [87]. Players' assessment of believability is primarily impacted by cultural affects and their skill level in the game [88]. This makes it difficult to quantify the human-likeness of NPCs or game levels. The approaches with the purpose of measuring the believability of AI can be classified in: adapted versions of the Turing test [89] and criteria rules proposed by experts [48,83,90]. The motivation for the latter methods is to reduce the subjectivity of tests. Although this criteria remains as subjective tests, they remove the subjectivity added by cultural affects and differences between novice and veteran players [88].

9.1. Section conclusions

We have identified that the most common manner to immerse players in a game is through believable NPC behaviors and scenarios. Regarding the believability concept, it is key to understand this is a subjective concept. Therefore, the perceived human-likeness of an NPC behavior or a game level depends on both the player and the game being played. Although player's traits impact how human-likeness is assessed, we found a lack of approaches intended to adapt how generated believable content is presented. Player typology is not considered when content is personalized. However, in the FlowAI framework, this factor is considered (see Fig. 4).

Also, we described in a broad manner the process to design and implement believable content trough AI techniques. In short, this is the process:

1. Understanding what makes game AI to appear human-like in a specific context.
2. Translate those abstract concepts about human-likeness to low-level characteristics related to the techniques intended to generate believable game AI.
3. Design a test to assess the level of human-likeness and evaluate our proposed approach by using it.

10. Summary and perspective

The starting point of this article was an overview on The Theory of Flow. We focused on explaining its main constructs and how they have been successfully used in many different applications. Then, in Section 4 we conducted an exhaustive analysis on approaches aimed to adapt flow elements to a video game context. Furthermore, in Section 5 we argue that immersion is a precursor of flow. More precisely, we aimed our attention on how AI techniques can facilitate immersion by tailoring gameplay elements.

With this theoretical foundation, we define a framework of flow for game AI in Section 6. Our framework, FlowAI, unifies the key elements of preexisting approaches that better suit a game AI context. Moreover, in contrast to others, it is structured as a generic player-centered game. This approach let us describe in more detail the operational function of each module and how they should drive game AI techniques; which has the intention of facilitating the players' achievement of flow. From our framework standpoint, we proceed to survey the game AI subareas that can contribute to foster flow: player modeling, Dynamic Difficulty Scaling (DDS) and immersion.

First, in Section 7, we described the essential elements of a player model that would fit our flow framework. Then, we proceeded to analyze the literature on this research area. The main issue we found is a generalized use of lab settings in the evaluation methods. This may bias reported results. Players should be able to play the game with their preferred setup and time.

Later, in Section 8 we presented a review of the literature including all works intended to maximize player's enjoyment by tailoring gameplay elements with an impact on a game's level of difficulty. Also, we concluded that the dynamic adaptation of difficulty is an optimization problem with the aim of minimizing the difference between the desired and perceived level of difficulty. Additionally, we believe that it would be valuable to perform more research in order to try to determine whether the reported correlation [35,47] between the perceived level of difficulty and an objective performance metric can be generalized.

Lastly, in Section 9 we give a brief overview about believable game AI and how it has a strong impact on driving players into a deeper level of immersion and, ultimately, into a flow state. Since believability is a subjective concept, we wonder if there would be a significant positive impact on the player's experience if we personalize the human-likeness of the game AI, that is, presenting the player an adapted human-like AI that better suits her player typology.

We believe that this article contributes with a comprehensible and formal analysis of the player-centered area, which will assist readers in the design, implementation and evaluation of effective player-centered games. Moreover, the FlowAI framework, despite being founded on purely theoretically basis, may influence researchers on the player-centered game AI research area, motivating new directions and research problems.

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